

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

First course Parasitology - Protozoa

Lecture I: General Parasitology

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Introduction

The parasites is an organism that lives (resides) on or in any living body called the host (it can be humans, animals, insect, and snail)

There are three main classes of parasites:

That can cause disease in human body, and they are: **endoparasites** which are helminths, protozoa, and the third one is **ectoparasites** (outside the human body) like insects, snail which is a small living organism (the parasite lives in it to complete the life cycle and for growth and transmission of the parasites).



Parasitology is the science which deals with the study of medical parasites, morphology, transmission with human diseases. A parasite is a living organism (eukaryotic) which is adapted to live in or on another living organism called the host. It also lives at the expense of the host, obtains its food and shelter from it.

There are two groups of animal parasites are of major importance in medical parasitology. **Protozoa and helminths** (metazoa), each constitutes different species according to their location in human body.



The most important characteristics:

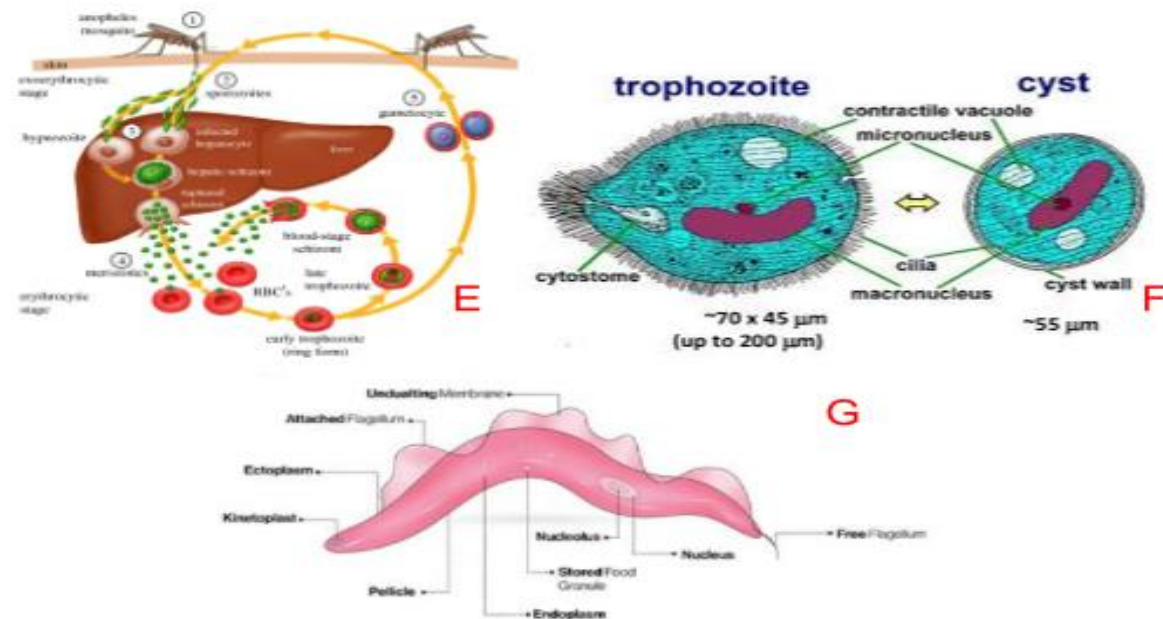
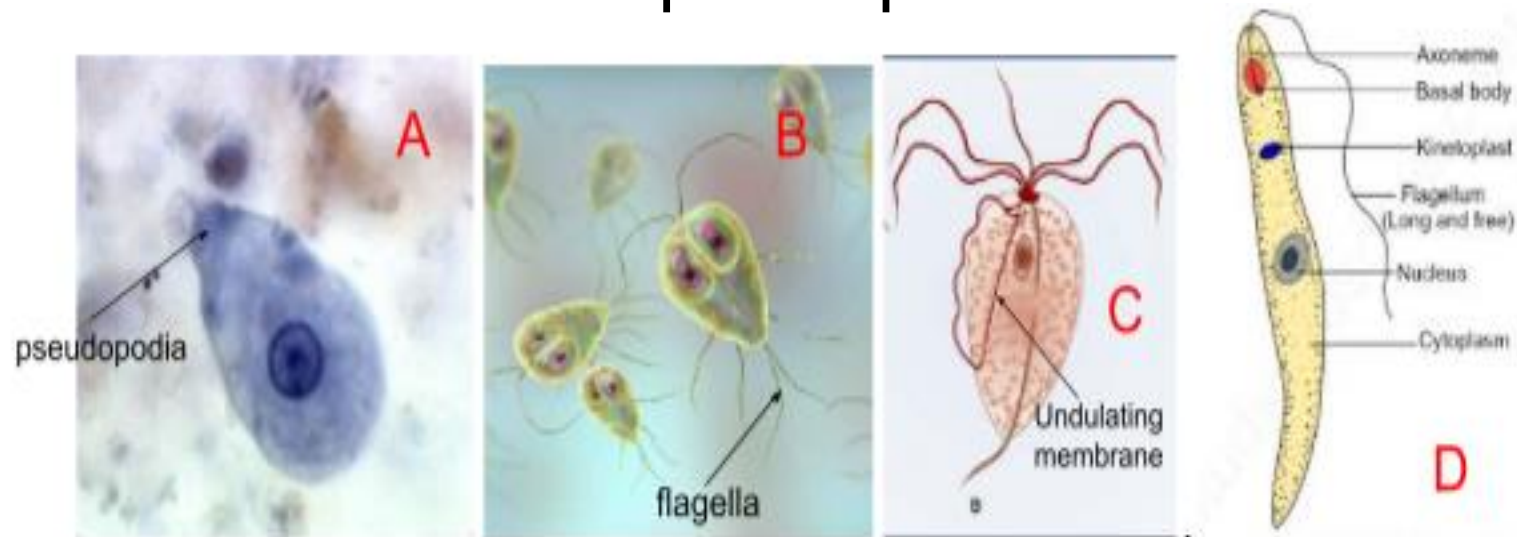
Protozoa:

Unicellular, the body consists of a single cell , e.g. ***Entamoeba spp***, ***Giardia lamblia***, ***Trichomonas spp.***, ***Trypanosoma spp.***, ***Leishmania spp.*** and ***Plasmodium spp.***

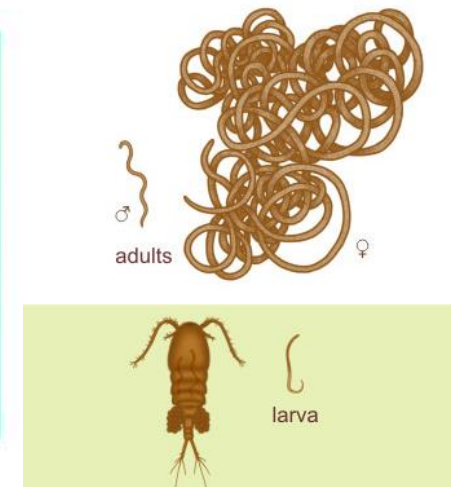
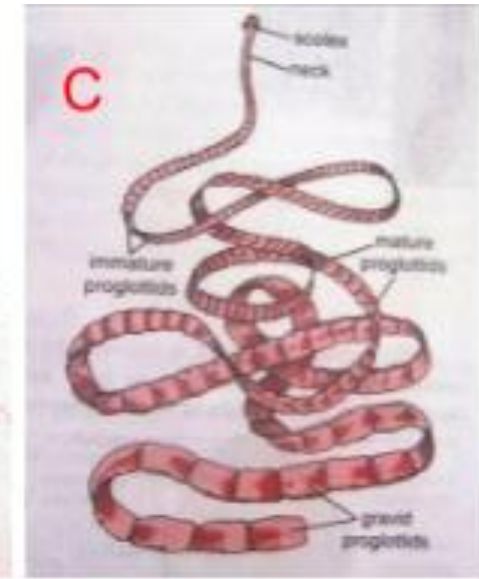
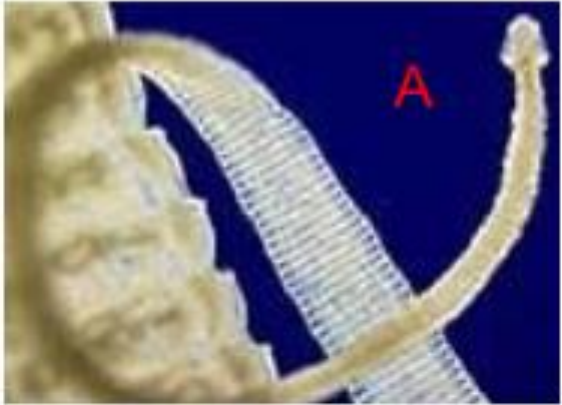
Helminths:

Multicellular, the body consists of more than one cell, e.g. ***Taenia saginata***, ***T. Solium***, ***E. Granulosus*** and ***H. Nana***.

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Examples of helminths





Protozoa have a complete cellular organelle as nucleus uni nucleus like trophozoites of ***Entamoeba Histolytica*** and multi nucleus like trophozoite of ***Giardia Lamblia***, ***Balantidium Coli***, and have mitochondria, endoplasmic reticulum and Golgi apparatus. Each protozoa perform all life activity.

Reproduction: reproduction of parasitic protozoa by sexual or asexual. The most common type of asexual multiplication is binary fission. Asexual reproduction as in the *Amoeba*, *Giardia*.

Some species of protozoa reproduce by both sexual and asexual methods like *plasmodium* in malaria.

- Asexual: schizogony in human body due to infection by *Plasmodium* (malaria).
- Sexual: sporogony or Gametocytes (male & female) in female anopheles, incorporation of the two cells (male and female) are called zygotes as in *Plasmodium species* (malaria) in female anopheles.

Locomotion:

Some species of protozoa are motile and this is possible by one of the following:

1. **Pseudopodia:** they are present in *Amoeba* (***Entamoeba Histolytica***)
2. **Cilia:** these are short like hair, they cover the whole body as in ***Balantidium Coli***.
3. **Flagellate:** these are one or few processes which are longer than cilia e.g. ***Giardia Lamblia*** and ***Trichomonas species*** (*T. Vaginalis*).
4. **Undulating membrane:** One of the undulating membrane are present in ***Trichomonas spp.***, ***Trypanosoma spp.***, and ***Leishmania spp.***



Respiration:

most free living protozoa respire aerobically by the contractile vacuole which liquid wastes are excreted by diffusion.

Nutrition: this is accomplished by one of the following modes in protozoa:

1. **saprophytic:** nutrients are obtained by absorption of simple molecules through the cell wall.
2. **Holozoic:** the protozoa engulf whole cells through a mouth then digestion occurs with food vacuoles in the endoplasm.

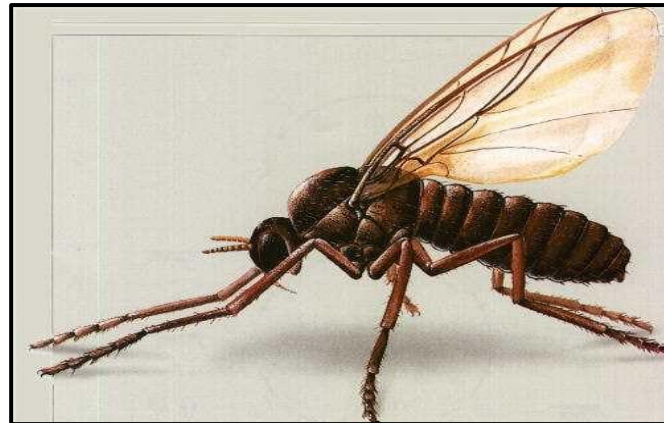
TYPES OF PARASITES

There are different types of parasites. The most important types are:

Endoparasite: This type of parasite lives inside the human body host and causes diseases.



Ectoparasite: These types of parasites live outside the human body host e.g. insects, ticks, snails and mites. They multiply and develop in the insect host or snail host to complete the life cycle.



Obligatory parasite:

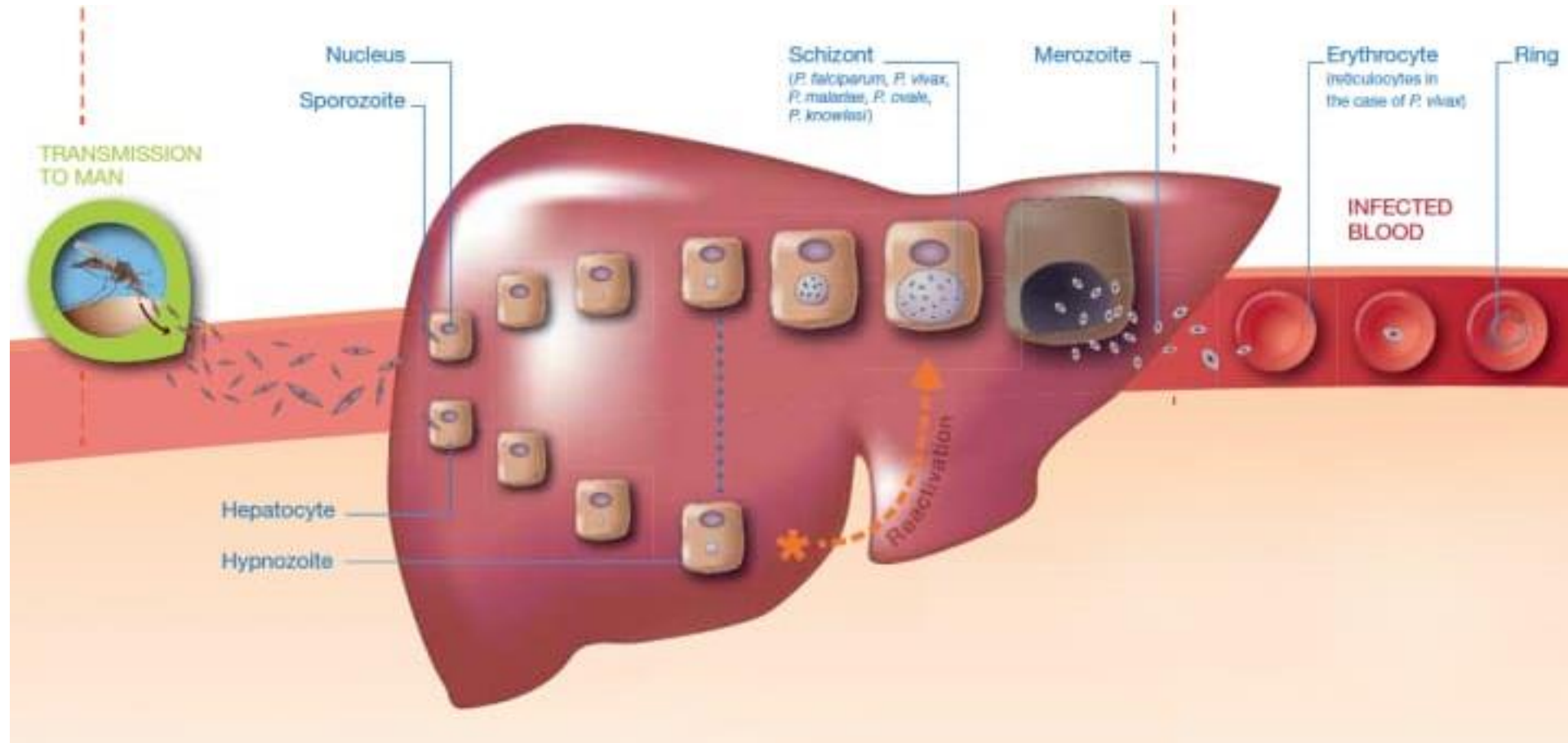
1. Obligate intracellular: This type can only live in the human body host and cannot survive without it. It exhibits the parasitic mode of life and is completely adapted for parasitic existence

- ***Plasmodium spp.*** (malaria) it is obligate intra baranchymal cell in cell of human liver,
- ***Toxoplasma gondii*** it is obligate in reticular endothelial cell (REC)

2. Obligate in human tissue:

- *Entameoba histolytica* the trohozoite infet the tissue of large intrsine
- *Schistosoma* species grow and develop in human liver to male and female

Accidental parasite: These parasites infect the human body accidentally through inhalation e.g. eggs of *Echinococcus granulosus* and eggs of *Enterobius vermicularis*.



Plasmodium (malaria)

Zoonosis

It is introduced by Rudolf Virchow in 1880 to include the disease shared by human and infected animal (naturally transmitted from infected animal to human body) like:

- **Helminths zoonosis:** e.g. hydatidosis (*Echinococcus granulosus*) and taeniasis (*Taenia saginata* and *T. Solium*).
- **Protozoa zoonosis:** e.g. *Toxoplasma gondii* causes toxoplasmosis, *Leishmania spp.* causes leishmaniasis and *balantidium coli* causes balantidiasis.

Source of infection

There are different sources of infections, and these may be:

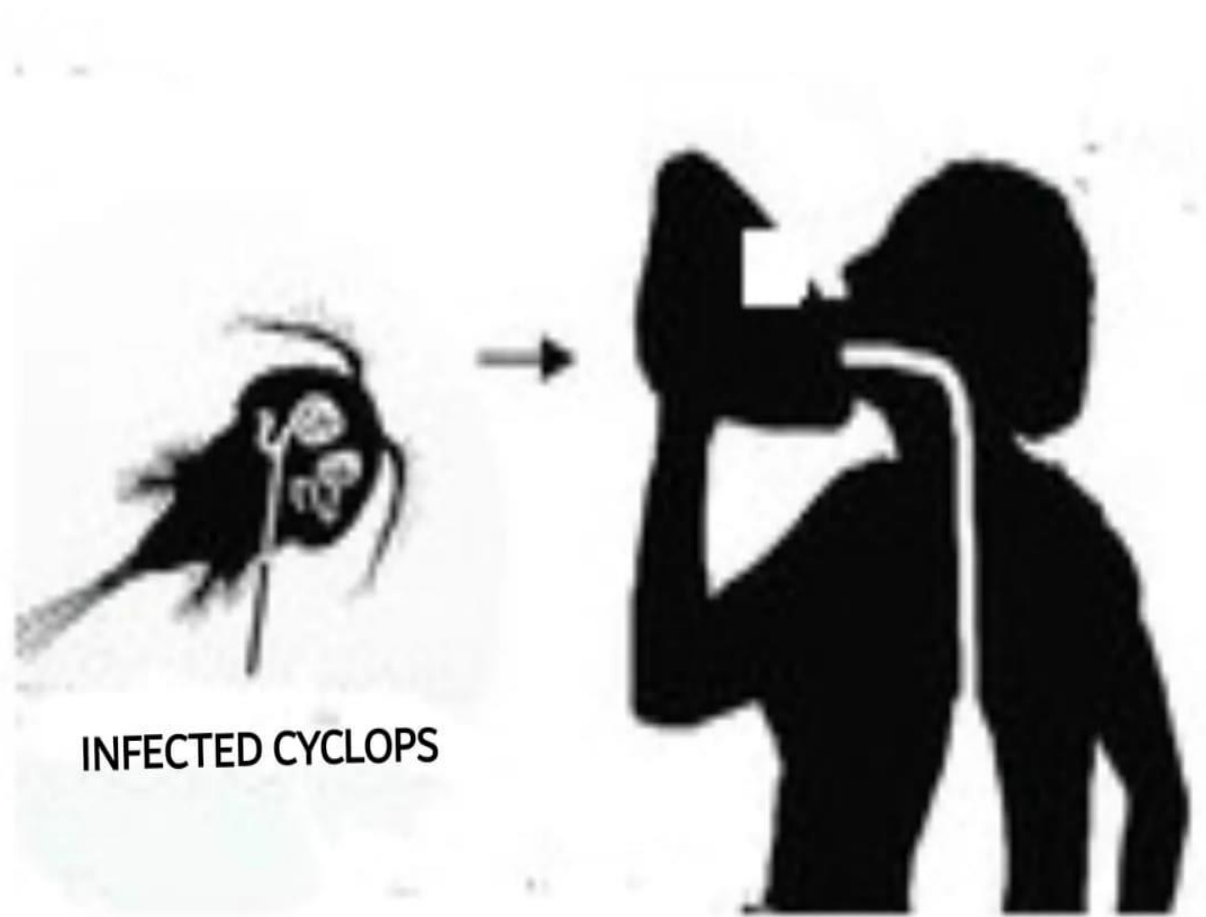
1. **Anthropophilic:** The source of infection is patients or carrier e.g. *Trichomonas vaginalis*
2. **Zoophilic:** The source of infection is infected animals as an intermediate host or a reservoir host e.g. *Balantidium coli*, and *Trichinella spiralis*.
3. **Geophilic:** The source of infection is the infected or contaminated soil e.g. *Hookworm* or contaminated water e.g. cercaria of *schistosoma*.

Transmission and mode of infection

There are two ways of transmission of parasite:

1. **Direct transmission:** This the way in which the infective stage of the parasitic disease invade the human body directly through skin contaminated by infected stage which is present in animals, soil, or from patients or human carriers, the infective stage may penetrate the skin directly or through sexual intercourse. Protozoa infection may be transmitted directly by a person to person when contacted e.g. by sexual intercourse like **vaginal trichomoniasis** and also by kissing in the case of ***Trichomanas tenax***.

2. **Indirect transmission:** The parasitic stage are transmitted from source of infection to the human body indirectly through ingesting, inhalation due to contamination clothes, open skin wounds, and also contaminated finger, flies, food, fluid, and feces, called the (F5), and through drinking the water and contaminated with parasitic stage (infective stage) or with intermediate host like cyclops which carries infective larva with ***dracunculus medinensis*** (Guinea worm)



Infected cyclops in water drinking by human and causes dracunculiasis



Parasitic infection may be transmitted by one of these ways:

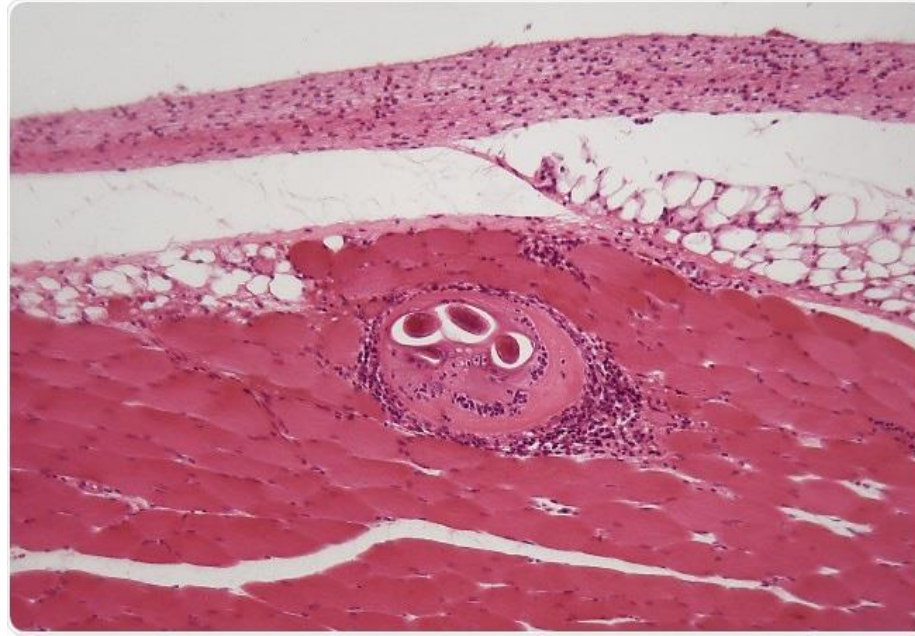
Ingestion : Ingestion of contaminate food or water with soil and dust containing the infective stage like intestinal protozoa: cyst stage of *E. histolytica* , *Giardia lamblia* and *Blantidium coli*, and helminthes like eggs of *Ascaris lumbricoides* , *Trichuris Trichiura*, ingestion or eating of infective fish containing active larva (meta-cercaria) of *paragonimus westermani* or *D. latum* and pork, cows, or cattle uncooked meat containing cysticercus larva to *Taenia saginata* or *T. Solium*, and cyst larva of *Trichinella spiralis* in meat of pork.

Cysticercus larva in cows



Taenia

Cysted larva in pork

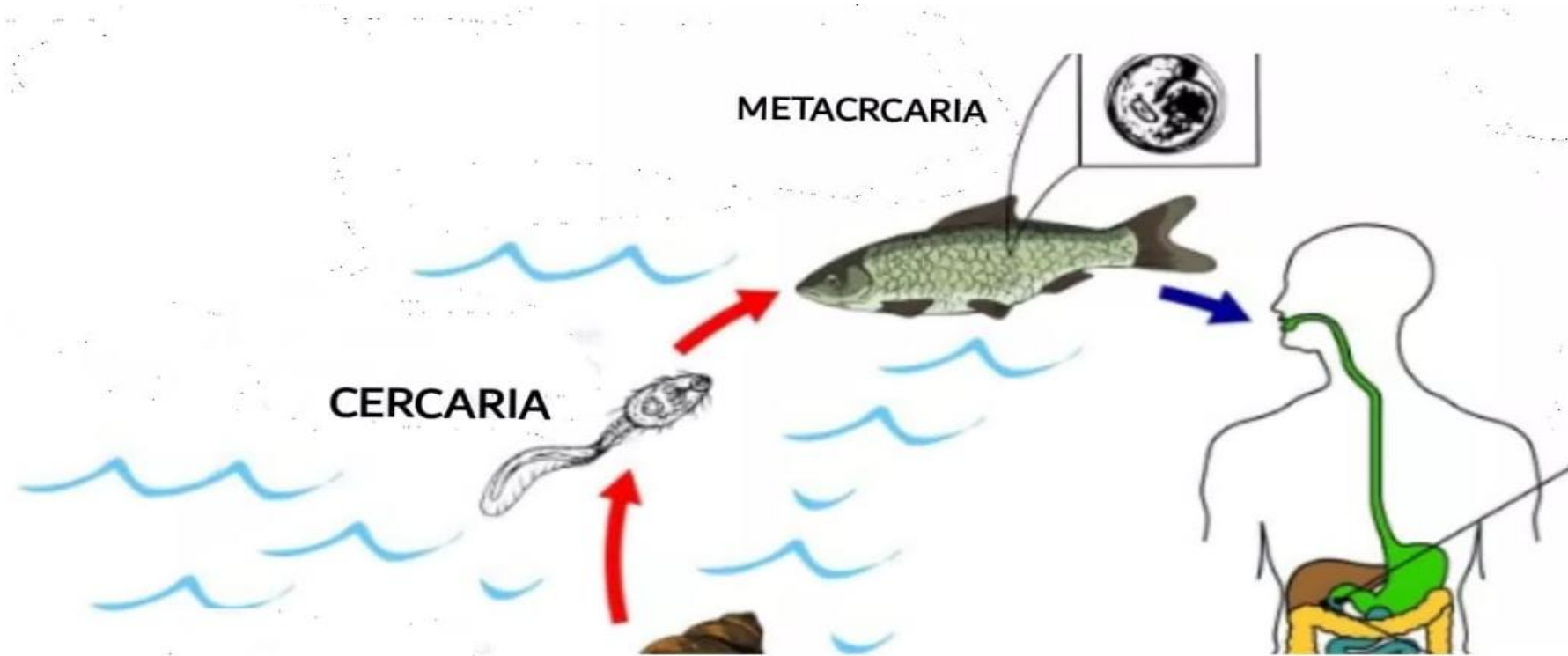


Trichinella spiralis

Active larva in fish



D. latum



- Metacercaria ingested by human and causes paragonimiasis
- Plerocercoid larva in infected fish ingested by human and causes diphyllobothriasis.

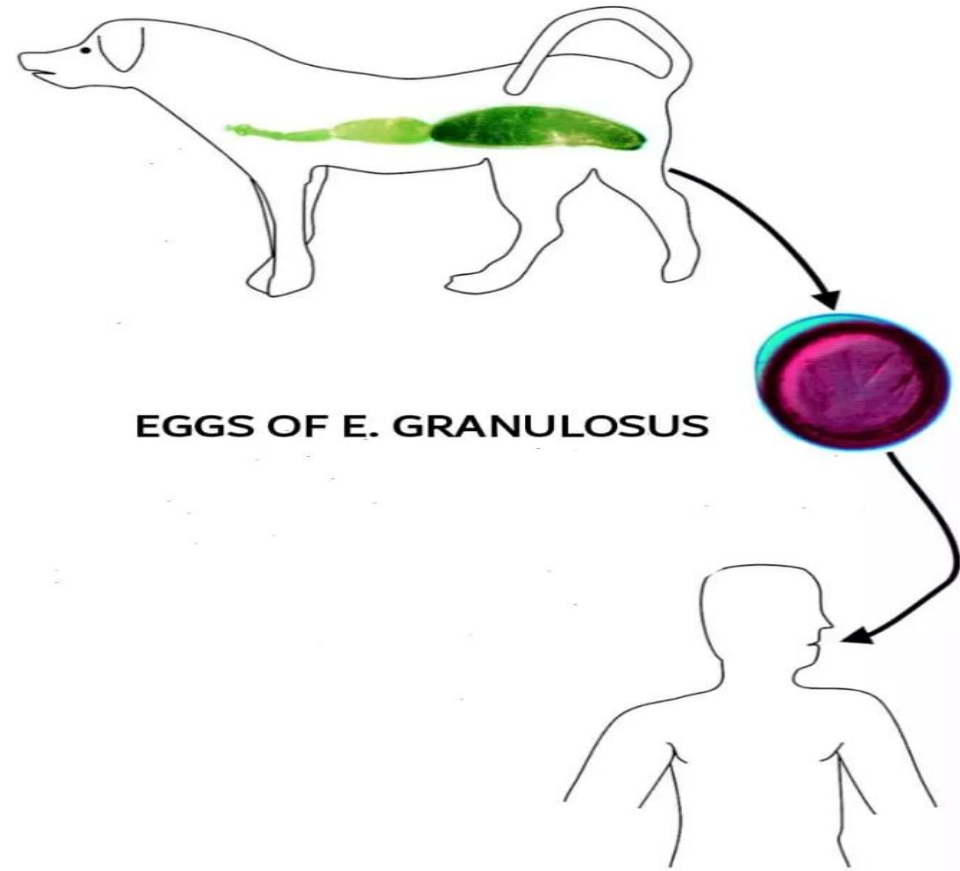


Inhalation of dust: carrying the infective stage e.g. eggs of *Enterobus vermicularis*, and *Echinococcus granulosus*

Skin: penetration of skin due to contact with infective stage e.g. filariform larva in soil to *hookworm* or *strongyloides stercoralis*, and when swimming or washing in water canal and streams the infective stage penetrate the skin or mucous membrane e.g. cercaria of *schistosoma*.



Eggs of *E. vermicularis* ingested or inhalation by human



Eggs of *E. Granulosus* ingested or inhalation by human



Cercaria of *schistosoma* penetrate human skin in water



Filariform larva of hook worm or *strongyloides stercoralis* in soil.



Vector: the parasitic stage are also transmitted from the source of infection to insect vectors as mechanical transmission or biological transmission by intermediate host.

Mechanical transmission: they are transmitted by leg, hair, joint appendages of insects from source of infection to human body directly or indirectly as contamination of the human skin, and food e.g. house flies transmit the infective stage like eggs of *Ascaris Lumericoides* from patient's stool or cyst stage of *Entamoeba Histolytica* (causes dysenteric amoebiasis) and cyst stage of *Giardia Lamblia* and causes giardiasis.



Transmission by vector



Biological transmission: some insects are intermediate hosts for protozoa and helminths or reservoir hosts. In which the parasitic stage in the vectors development only like helminths e.g. microfilaria of ***filaria worm*** (inactive microfilaria development to active microfilaria) in culex or aedes mosquito in which the parasitic stage multiply (growth) and development in vectors like protozoa e.g. ***Leishmania*** in sand fly.

Vector skin transmission: Many protozoal stages (infective stage) are transmitted by a special insect bite or deposit the feces. Bite transmission include:

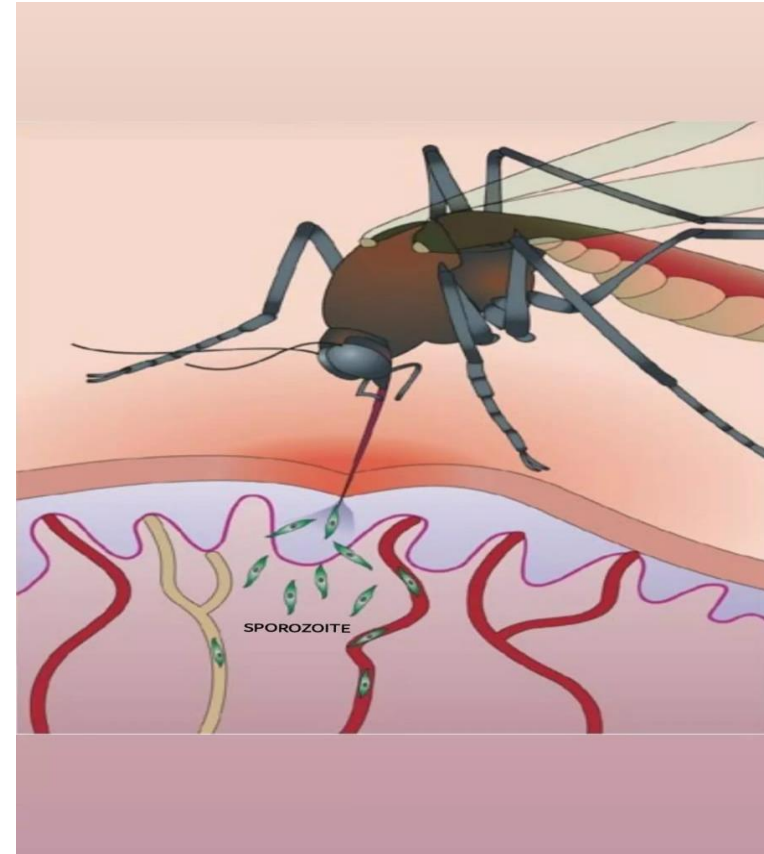
Malaria: when infected female anopheles bite and inject the sporozoite (infective stage)

Sleeping sickness: when tsetse fly bite and inject the metacyclic trypomastigote (infective stage).

Kala-azar: when sand fly bite and inject the promastigote (infective stage)



Reduviid deposit the feces with
(MCT) on human skin



Female anopheles inject sporozoite of malaria
in human skin



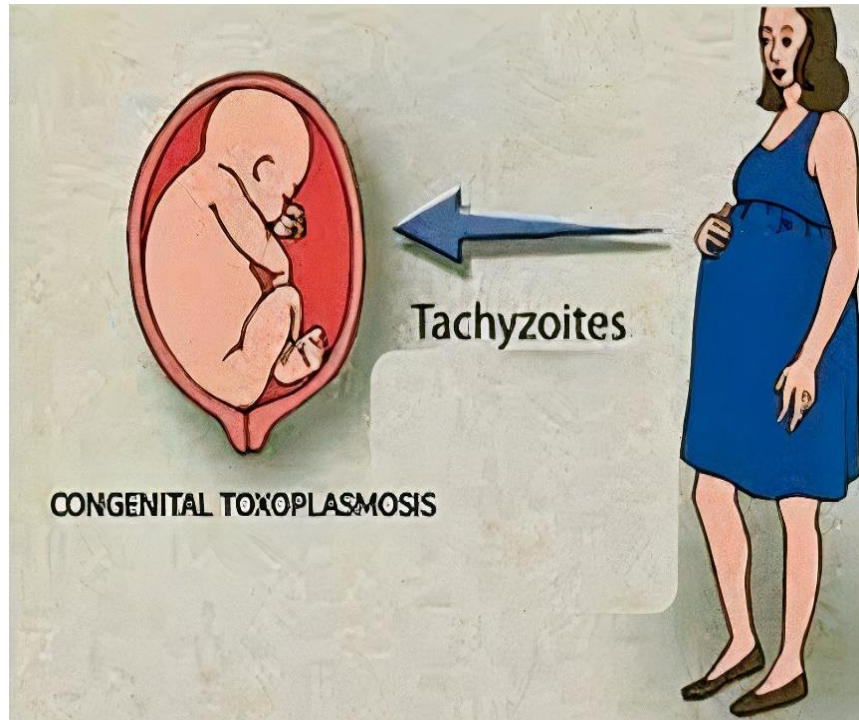
Insect inject the infective stage



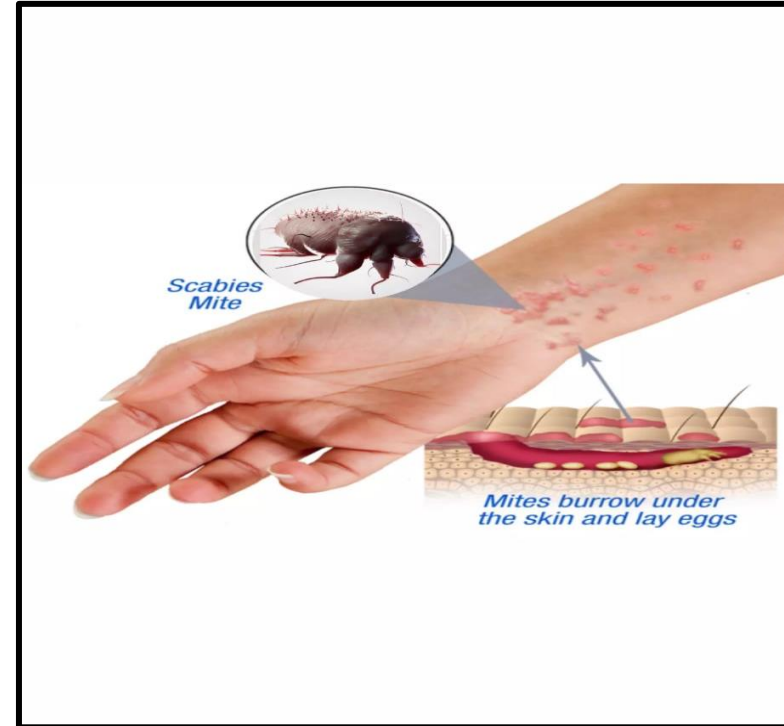
Protozoa infection: may be transmitted directly by a person to person when contacted e.g. by sexual intercourse like **vaginal trichomoniasis** and also by kissing in the case of ***trichomonas tenax***.

Spices of mite: like ***sarcoptes scabiei*** invade the adult in human skin and causes **scabies**.

Congenital (internal): some species of parasites are transmitted from infected mother to her fetus transplacentally this called congenital transmission disease like protozoa tachyzoite of ***toxoplasma gondii*** causes **congenital toxoplasmosis** and **congenital Chagas disease** for ***tryponasoma cruzi***.



Tachyzoites of *toxoplasma gondii* transmission, causes congenital toxoplasmosis



Sarcoptes scabiei invade human skin and causes scabies

Deposit the feces:

Chagas disease: when reduviid bugs put the feces on the skin with metacyclic trypomastigote (infective stage)

***Leishmania braziliensis*:** when sand fly put the feces on skin with promastigote (infective stage)

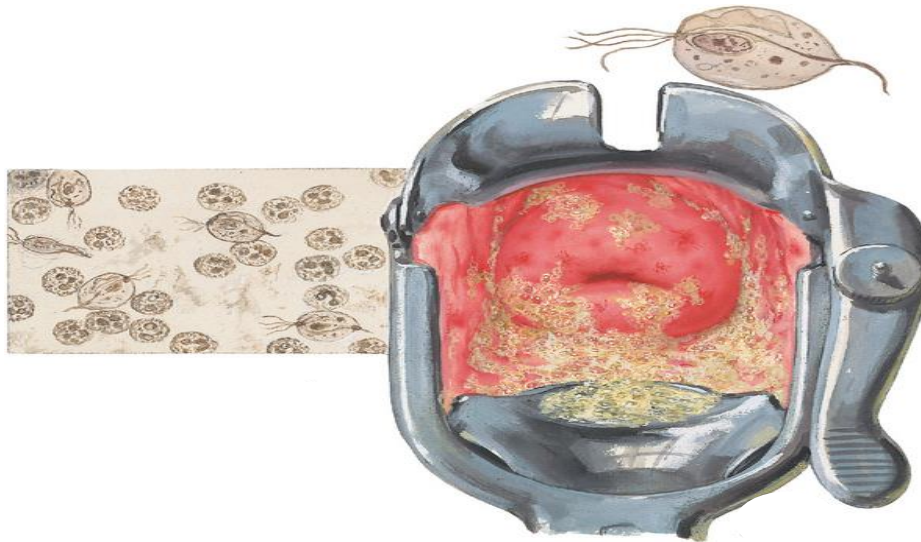


Insect deposit the feces with infective stage



Sexual transmission: The trophozoite stage of *trichomonas vaginalis* are transmitted through sexual intercourse and causes vaginitis or urethritis.

Intragenic transmission: Parasitic stages are transmitted with blood patients (transfusion and transplantation) to human body .



Trophozoite of *trichomonas vaginalis* invade the mucus membrane of vagina, male urethra and causes vaginitis or urethritis



Auto infection:

Auto infection or reinfection occurs when infective stage (eggs) are carried from perianal area to mouth of patient or larva invade the rectum through anus called external auto infection, and internal autoinfection e.g ***H. nana*** and larva of ***strongyloides stercoralis*** from internal and external auto infection in patients.



External autoinfection of *E. Vermicularis*



External autoinfection of *S. stercoralis*



Thank you for listening, any questions?